



King Abdul Aziz University

Faculty of Computing & Information Technology

Computer Science Department



CPCS 212 - APPLIED MATH FOR COMPUTING

Nonlinear Equation Secant Method

Instructor Dr. Ahmed Harbaoui

Outline

- Secant Method
- Examples
- Modified Secant Method
- Convergence Analysis

Newton's Method (Review)

*Assumptions: $f(x)$, $f'(x)$, x_0 are available,
 $f'(x_0) \neq 0$*

Newton's Method new estimate:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Problem:

$f'(x_i)$ is not available,
or difficult to obtain analytically.

Secant Method

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

if x_i and x_{i-1} are two initial points :

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{(x_i - x_{i-1})}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{\frac{f(x_i) - f(x_{i-1})}{(x_i - x_{i-1})}} = x_i - f(x_i) \frac{(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Secant Method

Assumptions :

Two initial points x_i and x_{i-1}
such that $f(x_i) \neq f(x_{i-1})$

New estimate (Secant Method) :

$$x_{i+1} = x_i - f(x_i) \frac{(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Secant Method

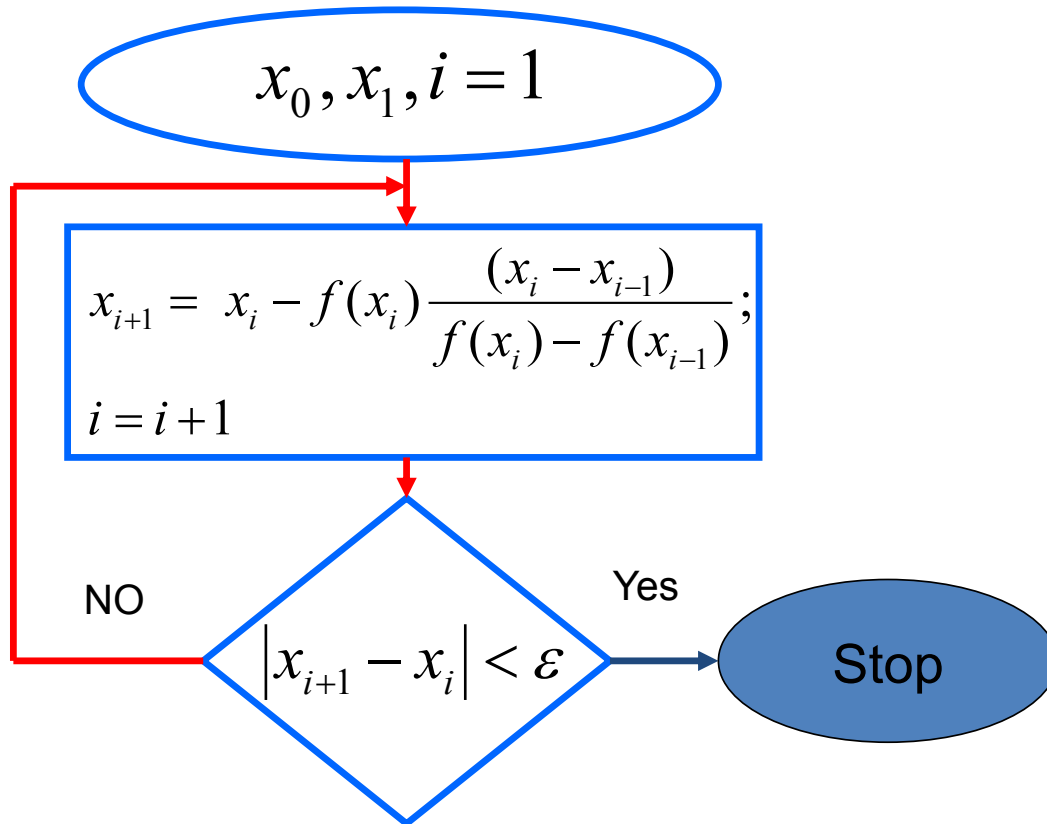
$$f(x) = x^2 - 2x + 0.5$$

$$x_0 = 0$$

$$x_1 = 1$$

$$x_{i+1} = x_i - f(x_i) \frac{(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Secant Method - Flowchart



Modified Secant Method

In this modified Secant method, only one initial guess is needed :

$$f'(x_i) \approx \frac{f(x_i + \delta_i x_i) - f(x_i)}{\delta_i x_i}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{\frac{f(x_i + \delta_i x_i) - f(x_i)}{\delta_i x_i}} = x_i - \frac{\delta_i x_i f(x_i)}{f(x_i + \delta_i x_i) - f(x_i)}$$

Problem : How to select δ_i ?

If not selected properly, the method may diverge.

Example

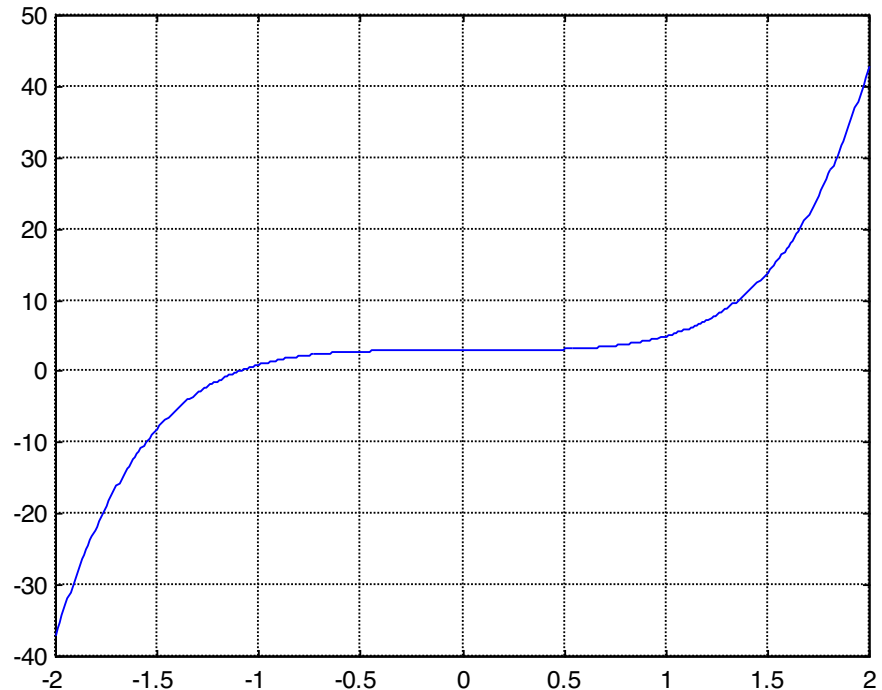
Find the roots of :

$$f(x) = x^5 + x^3 + 3$$

Initial points

$$x_0 = -1 \text{ and } x_1 = -1$$

with error < 0.001



Example

$x(i)$	$f(x(i))$	$x(i+1)$	$ x(i+1)-x(i) $
-1.0000	1.0000	-1.1000	0.1000
-1.1000	0.0585	-1.1062	0.0062
-1.1062	0.0102	-1.1052	0.0009
-1.1052	0.0001	-1.1052	0.0000

Example

Suppose $f(x) = x^4 - 5$, which has a solution $x^ = \sqrt[4]{5} \approx 1.5$*

Choose $x_0 = 1$ and $x_1 = 2$ as initial approximations

Next we have that $y_0 = f(1) = -4$ and $y_1 = f(2) = 11$

We may then calculate x_2

Convergence Analysis

The rate of convergence of the Secant method is super linear

$$\frac{|x_{i+1} - r|}{|x_i - r|^\alpha} \leq C, \quad \alpha \approx 1.62$$

r : root x_i : estimate of the root at the i^{th} iteration.

It is better than Bisection method but not as good as Newton's method.

Comparing Convergence Methods

Consider the function:

$$f(x) = x^3 - x - 2$$

We want to find its root using 3 numerical methods until the error is less than 10^{-6}

- Bisection Method $a=1.2$ and $b=1.8$
- Newton's Method $x_0 = 1.5$
- Secant Method $x_0=1.2$ and $x_1 =1.8$